The Andrew and Erna Viterbi Faculty of ELECTRICAL & COMPUTER ENGINEERING

Control Robotics and Machine Learning Laboratory

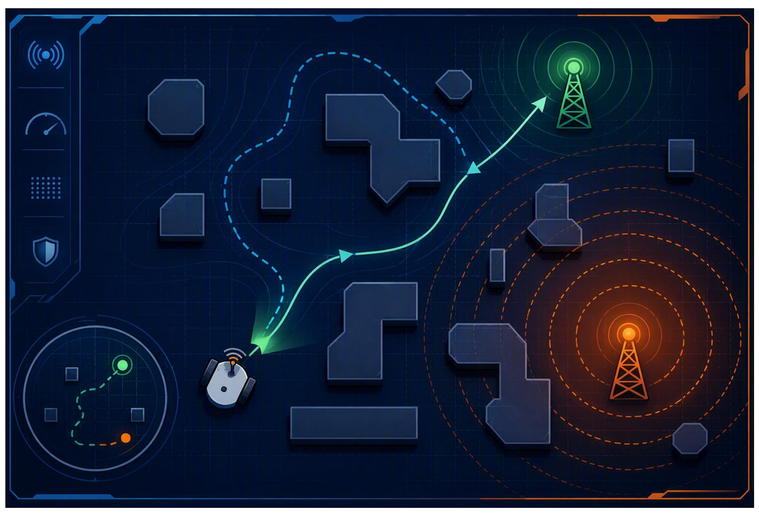
TECHNION Israel Institute of Technology

Adversarially Robust I-Bug Navigation

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Introduction

- **Minimalist Robotics:** Navigation without GPS, maps, or odometry.
- **The I-Bug Algorithm:** Uses only a contact sensor (for obstacles) and an intensity sensor (to find the target).
- **The Assumption:** The target broadcasts a symmetric, monotonically increasing signal.



Signal superposition: Target (green) vs. Pirate Tower (red).

The Adversarial Threat

- **Signal Superposition:** Real-world signals physically add up.

$$S(p,t) = m(p - p_t) + e(t)$$

- **Memory Poisoning:** A "pirate" tower transmits intermittent spoofed signals ($e(t)$), creating false intensity peaks.
- **The Trap:** The robot saves a fake, artificially high threshold (i_H) upon hitting a wall, permanently trapping it in infinite loops when the jamming stops.

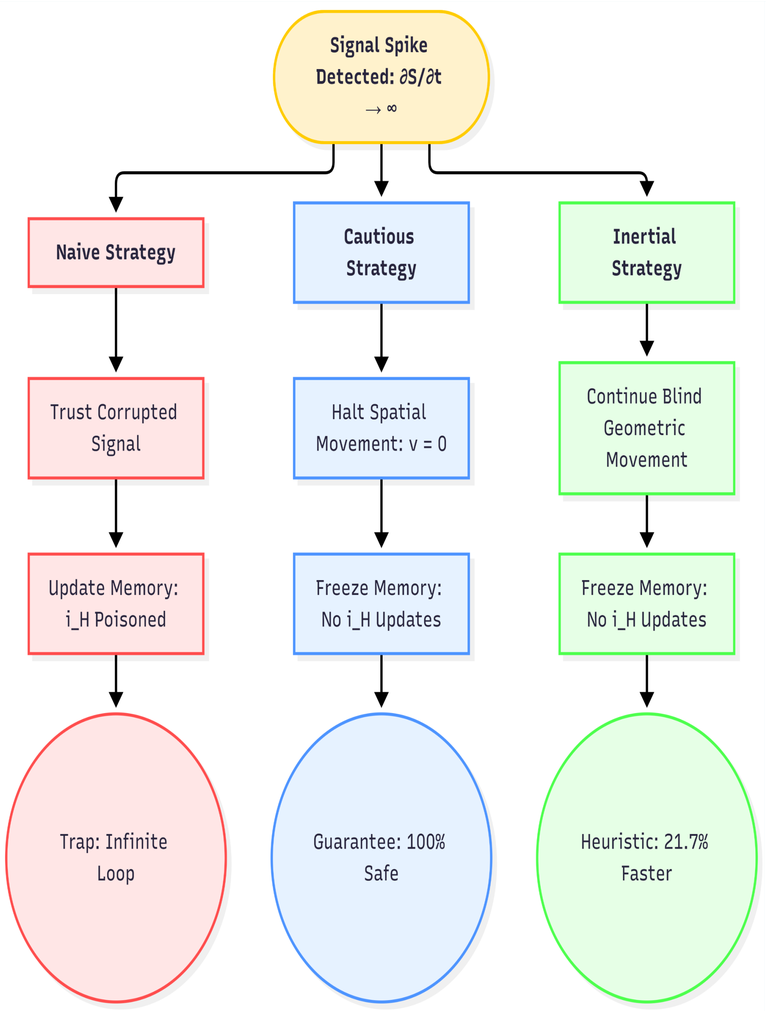
Goals

- **Autonomous Detection:** Identify attacks mathematically via temporal signal spikes ($\partial S/\partial t \to \infty$).
- **Robust Control**: Develop software-level behavioral strategies to survive attacks without new hardware.
- **Empirical Validation:** Quantify the safety vs. efficiency trade-off.

Simulation Setup

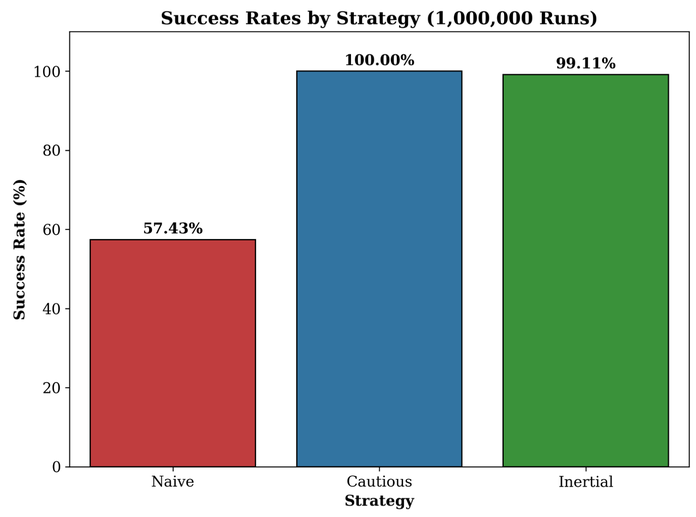
- **Engine**: Continuous-resolution 2D kinematics in Julia.
- **Scale:** 1,000,000 procedurally generated randomized environments.
- **Evaluation:** Paired execution (all 3 robots face the exact same layout and jamming cycles concurrently).

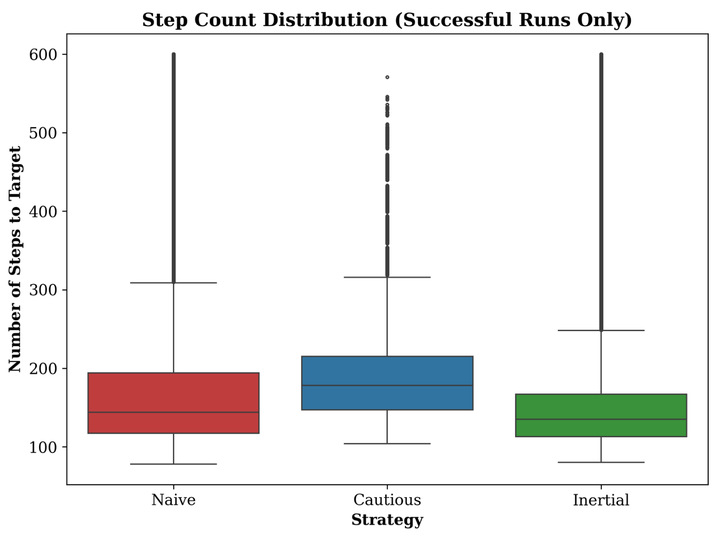
The Proposed Strategies (Algorithm)



Reliability vs. Efficiency

- **Naive**: Catastrophic failure (42.57% trap rate).
- **Cautious**: Flawless 100% success (Mathematically guaranteed path-preservation).
- **Inertial**: Highly successful heuristic (99.11% success).
- **The Efficiency Trade-off**: Cautious pays a temporal penalty (189 steps). Inertial saves time (147.9 steps) →21.7% efficiency improvement.





Mathematical Guarantee (Cautious Strategy)

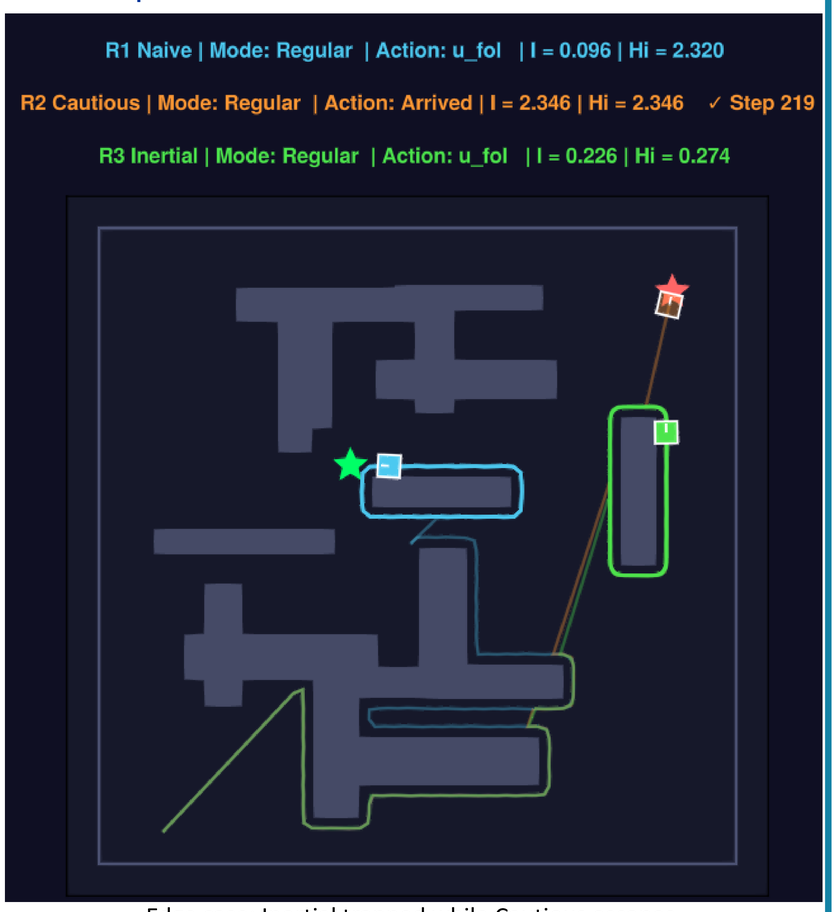
- **Time-Deletion:** Halting spatial movement (v=0) and freezing memory during an anomaly effectively "erases" the corrupted time segments from the robot's local timeline.
- **Trajectory Reduction:** This pause perfectly maps the robot's physical trace back to a sterile, interference-free environment

$$\gamma_{cautious} \equiv \gamma_{ideal}$$

- **Inherited Proof:** By preserving the exact geometric path, the strategy strictly inherits the absolute, finite-time convergence guarantees proven in the original Taylor & LaValle (2014) framework.

Edge Case: Cyclic Synchronization

- Why did the Inertial strategy fail in 0.89% of cases?
- Blind movement suspends local-maxima checks.
- In specific non-convex geometries, it bypasses valid departure points, creating topological resonance.
- As seen below: Inertial robot (green) trapped in an infinite loop, while Cautious (orange) safely escapes.



Edge case: Inertial trapped while Cautious escapes.

Conclusions

- Minimalist robots can achieve adversarial robustness purely through logical memory management and anomaly detection.
- This establishes a clear engineering choice: use the Cautious strategy for absolute mathematical safety, or the Inertial heuristic for maximum operational speed.